PHASE-FILTERED RAMANUJAN SUMS AND THE SPECTRAL GATE

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Abstract. For primitive-root primes, the autocorrelation $R(\ell)$ of the base repetend can be interpreted as a phase-filtered Ramanujan sum, where the filter is the cross-spectral function G of the bin partition evaluated along the multiplicative orbit of the base.

The continued fraction expansion of p/b has a tail determined by $r = p \bmod b$, partitioning primes into $\varphi(b)$ spectral classes. Within each class, the coarse bin pattern (the number of large bins, $r-1 = (p-1) \bmod b$) is determined by the class. The continued fraction tail organizes the coarse phase geometry observed in the autocorrelation data.

1. Introduction

This paper expresses the autocorrelation as a phase-filtered Ramanujan sum.

Given a prime p not dividing b with repetend $(d_0, d_1, \ldots, d_{L-1})$ for $1/p$ in base b, the *autocorrelation of the repetend* at lag ℓ is

$$R(\ell) = |\{j \in \{0, \ldots, L-1\} : d_j = d_{(j+\ell) \bmod L}\}|,$$

the number of positions j at which the repetend digit equals the digit ℓ positions later (cyclically).

The classical Ramanujan sum [3]

$$c_q(n) = \sum_{\substack{k=1 \\ \gcd(k,q)=1}}^{q} e^{2\pi i k n/q}$$

assigns a uniform weight to each coprime residue. For a prime $q = p$, $c_p(n) = -1$ for all $n \not\equiv 0 \pmod{p}$, and $c_p(0) = p-1$. The Ramanujan expansion of arithmetic functions [4] connects these sums to the Riemann zeta function through the identity

$$\sum_{q=1}^{\infty} \frac{c_q(n)}{q^s} = \frac{\sigma_{1-s}(n)}{\zeta(s)}.$$

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The autocorrelation $R(\ell)$ of the base repetend, derived in the autocorrelation formula [1], also sums over residues modulo p. But unlike c_p, it does not treat all nonzero lags equally. Some lags produce partial constructive interference ($R(\ell) > 0$) while others produce exact cancellation ($R(\ell) = 0$). The mechanism governing this discrimination is the *phase geometry* of the digit bins.

This paper identifies $R(\ell)$ as a phase-filtered Ramanujan sum, characterizes the filter through the continued fraction expansion of p/b, and classifies all primes into finitely many spectral classes by their residue modulo b.

2. The Phase Filter

Definition 1. The *phase filter* at lag ℓ is the ratio

$$\Gamma(\ell) = \frac{|\Sigma_0(\ell)|}{M(\ell)},$$

where $\Sigma_0(\ell) = \sum_k G(k, -kb^{-\ell} \bmod p)$ is the cross-spectral sum from the autocorrelation formula [1] and $M(\ell) = \sum_k \sum_d |\widehat{\mathbf{1}}_{B_d}(k)|\,|\widehat{\mathbf{1}}_{B_d}(-kb^{-\ell} \bmod p)|$ is the corresponding magnitude sum.

The phase filter satisfies $0 \leq \Gamma(\ell) \leq 1$. When $\Gamma = 1$ (all phases aligned), the cross-spectral sum achieves its maximum and $R(\ell)$ is large. When $\Gamma = 0$ (complete cancellation), $R(\ell) = 0$ regardless of the magnitudes.

Observation 2. For a prime p where b is a primitive root (verified computationally for all primes $p \leq 10000$ in bases $b \in \{3, 6, 7, 10, 12\}$):

(1) $\Gamma(0) = 1$ (self-lag: perfect alignment).
(2) For $\ell \neq 0$: $\Gamma(\ell) \in \{0\} \cup [c_{\min}, c_{\max}]$ where $c_{\min} > 0$. The filter is either exactly zero or bounded away from zero.
(3) The set $\{\ell : \Gamma(\ell) > 0,\ \ell \neq 0\}$ has cardinality determined by $r = p \bmod b$.

The phase filter converts the uniform Ramanujan sum $c_p(\ell) = -1$ into the selective autocorrelation $R(\ell)$: it "gates" which lags contribute to the spectral structure and which are suppressed by destructive interference.

3. The Continued Fraction Control

The bin pattern of the digit function $\delta(r) = \lfloor br/p \rfloor$ is governed by the continued fraction expansion of p/b.

Proposition 3. *The continued fraction* [5] *of* p/b *has the form*

$$\frac{p}{b} = [q,\ a_1, a_2, \ldots] = \left[q,\ \mathrm{CF}\left(\frac{b}{r}\right)\right],$$

where $q = \lfloor p/b \rfloor$ *and* $r = p \bmod b$*. The tail* $\mathrm{CF}(b/r)$ *depends only on* r *and* b*, not on* p*.*

Proof. The Euclidean algorithm: $p = qb + r$, so $p/b = q + r/b$ and the continued fraction continues with b/r. □

Since b is fixed, the tail of the CF is determined by r alone. In base 10, the coprime residues $r \in \{1, 3, 7, 9\}$ produce four distinct CF tails:

$r = p \bmod 10$	CF tail	Bin pattern
1	$[10]$	uniform (0 large bins)
3	$[3, 3]$	2 large bins
7	$[1, 2, 3]$	6 large bins
9	$[1, 9]$	8 large bins

Table 1. Spectral classes in base 10. The CF tail determines the bin pattern and hence the phase geometry of G.

4. Spectral Classes

Definition 4. Two primes p_1, p_2 (not dividing b) belong to the same *spectral class* if $p_1 \equiv p_2 \pmod{b}$.

Within a spectral class, all primes share the same CF tail and the same coarse bin pattern ($(p-1) \bmod b$ large bins). The number of constructive lags depends on p itself ($p - b - 1$ for primitive-root primes), not on the spectral class alone.

Theorem 5. *In base* b*, there are* $\varphi(b)$ *spectral classes, where* φ *is Euler's totient function. In base* 10*, there are four:* $r \in \{1, 3, 7, 9\}$*.*

Proof. The admissible residue classes are exactly the classes $r \pmod{b}$ with $\gcd(r, b) = 1$, since primes not dividing b must lie in such classes. There are $\varphi(b)$ such classes by definition of Euler's totient function. In base 10, these are $r \in \{1, 3, 7, 9\}$. □

The spectral class of p determines:

- the number of large bins: r,
- the CF tail: $\mathrm{CF}(b/r)$,
- the coarse bin geometry used in the observed filter patterns.

5. Ramanujan Sums and the Autocorrelation

The Ramanujan sum $c_p(\ell)$ and the autocorrelation $R(\ell)$ are both sums over $(\mathbb{Z}/p\mathbb{Z})^*$ involving additive characters. Their relationship can be stated precisely.

Proposition 6. *For a prime p where b is a primitive root:*

$$R(\ell) = \frac{1}{p}\sum_{k=0}^{p-1} G(k, -kb^{-\ell} \bmod p),$$

while

$$c_p(\ell) = \sum_{k=1}^{p-1} e^{2\pi i k\ell/p}.$$

Both sums run over additive characters of $\mathbb{Z}/p\mathbb{Z}$. The Ramanujan sum weights each character uniformly ($e^{2\pi i k\ell/p}$). The autocorrelation weights each character by the cross-spectral function G, evaluated at a point determined by the multiplicative orbit of b.

The autocorrelation $R(\ell)$ is therefore a Ramanujan-type sum with a non-uniform weight function. The weight function is G, which encodes the digit-bin geometry. The Ramanujan sum is the special case where G is replaced by the trivial weight (all bins of size 1, i.e., the digit-partitioning case).

Corollary 7. *For digit-partitioning primes,*

$$G(k,k') = \begin{cases} p-1 & \text{if } k+k' \equiv 0 \pmod p, \\ -1 & \text{otherwise,} \end{cases}$$

and

$$R(\ell) = (p-1)\delta_{\ell 0}.$$

The phase filter passes only the self-lag, and the autocorrelation reduces to the identity contribution.

Proof. For digit-partitioning primes, every nonempty bin is a singleton, so

$$G(k,k') = \sum_{r=1}^{p-1} e^{-2\pi i(k+k')r/p}.$$

The standard nontrivial-character sum gives the displayed two-valued formula. Substituting into Proposition 6, the case $\ell = 0$ samples the antidiagonal $k' = -k$ and contributes $(p-1)$ at every k, so $R(0) = p-1$. For $\ell \neq 0$, exactly one term in the sum (the term $k = 0$) contributes $p-1$ and the remaining $p-1$ terms contribute -1, so the total is 0. $\square$

6. Toward a Modified Zeta Identity

The classical Ramanujan expansion connects $c_q(n)$ to $\zeta(s)$. If $R(\ell)$ can be expressed as a modified Ramanujan sum with a correction from G, a modified zeta identity may exist.

Define the *structural Ramanujan sum* at prime p:

$$c_p^G(\ell) = \frac{1}{p} \sum_{k=0}^{p-1} G(k, -kb^{-\ell} \bmod p).$$

For primitive-root primes, $R(\ell) = c_p^G(\ell)$ by the autocorrelation formula [1]. The generating function

$$L^G(s) = \sum_p \frac{c_p^G(\ell)}{p^s}$$

(summing over primes at fixed ℓ) is a Dirichlet series encoding how the phase-filtered autocorrelation distributes across primes.

Whether $L^G(s)$ has meromorphic continuation, a functional equation, or a relationship to $\zeta(s)$ analogous to the classical Ramanujan identity is an open question. The spectral class structure (Theorem 5) implies that L^G decomposes into contributions from $\varphi(b)$ residue classes, suggesting a connection to Dirichlet L-functions $L(s, \chi)$ for characters χ modulo b [6].

This direction is speculative. The series is defined only for primitive-root primes, and its analytic properties are not established.

7. Remarks

The spectral gate. The phase filter $\Gamma(\ell)$ acts as a gate: it admits certain Ramanujan lags into the autocorrelation and blocks others. The gate is controlled by the digit-bin geometry, which is controlled by the CF expansion of p/b. The gate is therefore a structural property of the interaction between the prime and the base, not a property of either alone.

For DP primes, the gate is maximally restrictive: only the self-lag passes. For non-DP primes, the gate opens selectively. The spectral class organizes the coarse bin geometry behind the observed gate patterns.

Connection to earlier work. The recursive digit resolution framework [2] studies how the interaction between a base grid and a prime grid creates structure at different resolution scales. The continued fraction expansion of p/b encodes these scales. The partial quotients are the resolution levels of the Moiré pattern between the two grids. Large partial quotients correspond to near-alignment (flat spectrum,

DP-like behavior). Small partial quotients correspond to misalignment (structured spectrum, rich autocorrelation).

The spectral gate is the formal mechanism by which this resolution interaction governs the coherence of the fractional field.

References

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